

CS 6220: Natural Language Inference Project Report

Team Members and Github Link

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Github Repository: <https://github.com/2025-F-CS6220/project-anliexps>

Project Overview

This project tackles Natural Language Inference (NLI) on the Adversarial NLI (ANLI) Round 2 dataset, a challenging benchmark designed to expose weaknesses in state-of-the-art NLP models. The goal is to classify the semantic relationship between premise-hypothesis pairs into three categories: entailment, neutral, or contradiction. The project systematically explores approaches ranging from traditional machine learning baselines to advanced transformer architectures.

The project achieved significant success, improving from a 33.7% baseline accuracy to 61.8% with DeBERTa-v3-large—a 28.1 percentage point absolute improvement representing 83.4% relative gain. This was accomplished through comprehensive exploratory data analysis revealing critical dataset characteristics (class imbalance, low word overlap, high semantic complexity), establishing traditional ML baselines (Logistic Regression, Random Forest, XGBoost achieving 35.6-38.7%), and progressive fine-tuning of transformer models (BERT-base: 43.1%, BERT-large: 44.7%, BERT-large+CoT: 45.4%). The best model demonstrates balanced performance across all three classes (F1: 0.5958-0.6416) with excellent generalization, as test accuracy (61.8%) exceeds validation accuracy (61.1%). Key insights include: (1) DeBERTa's disentangled attention architecture provides a 16.4% improvement over BERT for adversarial examples, (2) prompt engineering alone yields minimal gains (+0.7%), and (3) the adversarial nature of ANLI requires specialized fine-tuning beyond pre-training on simpler NLI datasets.

Input Data

Dataset: Adversarial Natural Language Inference (ANLI) Round 2

Source: HuggingFace datasets library

Size: 45,548 training samples, 1,000 validation samples, 1,000 test samples

Structure: Each sample contains:

- **Premise:** A context statement or passage

- **Hypothesis: A statement to evaluate against the premise**
- **Label: The relationship (0=entailment, 1=neutral, 2=contradiction)**
- **Reason: Human-provided explanation for the label**

Sample Records:

Example 1 (Entailment):

Premise: "The End Tour was the farewell tour of Black Sabbath, featuring Ozzy Osbourne, Tony Iommi and Geezer Butler. The tour concluded Sabbath's 40+ year career."

Hypothesis: "Some bands from Birmingham, UK have had careers extending beyond 4 sets of ten years."

Label: Entailment

Example 2 (Neutral):

Premise: "The National Democratic Party (NDP) is a Ghanaian political party, founded in October 2012."

Hypothesis: "The party is considering holding elections in 2020."

Label: Neutral

Example 3 (Contradiction):

Premise: "Richard Colson Baker, better known as MGK, released a mixtape in 2006. He was born April 22, 1990."

Hypothesis: "Richard Colson Baker released his first mixtape at age eighteen."

Label: Contradiction

Key Dataset Characteristics from EDA:

- **Text Length:** Premises average ~50 words (range: 10-250 words), hypotheses average ~10 words, informing the choice of 256-token max sequence length
- **Class Imbalance:** Training data shows imbalance with neutral (46.1%), entailment (31.8%), contradiction (22.1%)

- Low Word Overlap: Jaccard analysis reveals surprisingly low overlap (10-15%), with contradictions unexpectedly showing highest overlap (14.3%), demonstrating that simple word matching fails for NLI
 - Similarity Scores: TF-IDF cosine similarity moderate (0.4-0.6), optimal feature size ~15,000
 - Adversarial Design: ANLI was specifically crafted to fool BERT-like models through adversarial sentence pairs that exploit architectural weaknesses
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Problem

Task: Given a premise and hypothesis text pair, predict the semantic relationship as one of three classes:

- Entailment: The hypothesis logically follows from the premise
- Neutral: The hypothesis might be true given the premise, but isn't necessarily implied
- Contradiction: The hypothesis directly conflicts with the premise

Challenges: The ANLI R2 dataset is adversarially constructed to challenge modern NLP models through:

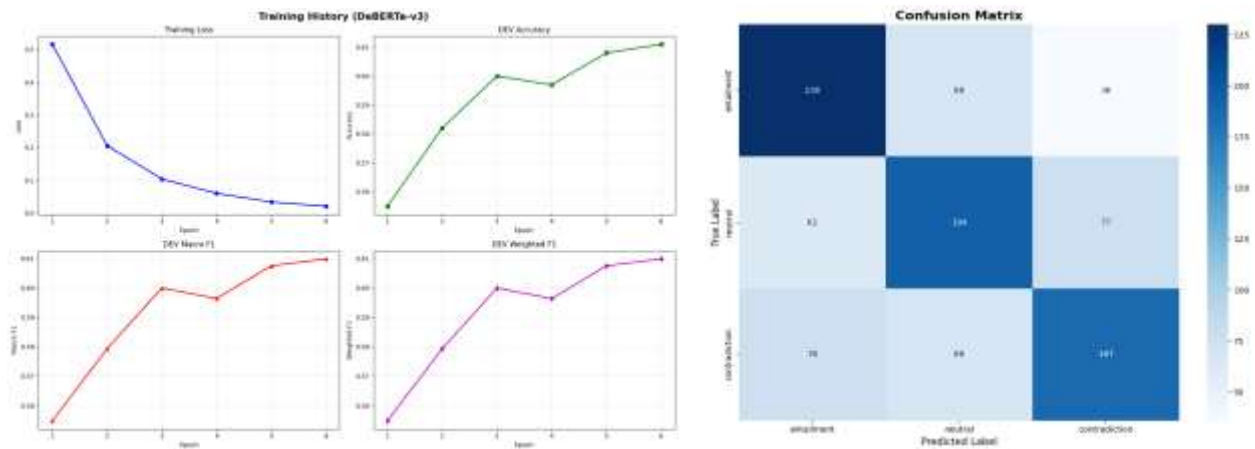
- (1) high semantic ambiguity with close similarity scores across classes,
- (2) class imbalance potentially biasing predictions,
- (3) requirement for deep contextual understanding beyond surface-level word matching, (4) deliberately adversarial examples designed to exploit BERT weaknesses, and
- (5) limited training data (45K samples) for complex 3-way classification.

The task requires models to handle negation, numerical reasoning, temporal logic, and subtle semantic distinctions.

Evidence of Success

Quality Measure Selection

Primary Metric: Accuracy - Selected because the test set maintains perfectly balanced class distribution (~333 samples per class), making accuracy a reliable and interpretable measure. The balanced test set ensures accuracy doesn't favor any particular class.



Training History: Model converges efficiently with training loss decreasing from 0.5 to near 0 over 6 epochs, while dev accuracy and F1 scores plateau at 0.61 (~61%), triggering early stopping to prevent overfitting.

Confusion Matrix: Balanced diagonal dominance (230, 194, 187 correct predictions per class) with relatively even error distribution across all class pairs, confirming no systematic bias toward any particular class despite training set imbalance.

Secondary Metric: Macro F1 Score - Ensures balanced performance across all three classes without bias toward any particular label, especially critical given the training set class imbalance (46.1% neutral). Macro F1 treats all classes equally regardless of support.

Comprehensive Results

Approach	Accuracy	Macro F1	Improvement
Baseline (DistilRoBERTa)	33.7%	0.242	-
Logistic Regression (TF-IDF)	35.6%	0.339	+1.9%
Random Forest (TF-IDF)	36.5%	0.245	+2.8%
XGBoost (TF-IDF)	38.7%	0.329	+5.0%
BERT-base	43.1%	0.427	+9.4%
BERT-large	44.7%	0.443	+11.0%
BERT-large + CoT Prompting	45.4%	0.450	+11.7%
DeBERTa-v3-large	61.8%	0.6177	+28.1%

Key Findings:

- Traditional ML provides solid baselines but plateaus at 38.7% (XGBoost), validating EDA prediction that low word overlap limits feature-based approaches
- BERT models show consistent improvement with scale (base: 43.1% → large: 44.7%) but plateau around 45%
- Prompt engineering offers minimal gain (+0.7%), indicating limitations of BERT architecture on adversarial examples
- DeBERTa-v3-large achieves breakthrough performance (+16.4% over BERT-large) through disentangled attention and enhanced mask decoder

Classification Report:				
	precision	recall	f1-score	support
entailment	0.6124	0.6527	0.6319	334
neutral	0.5862	0.6126	0.5991	333
contradiction	0.6284	0.5586	0.5914	333
accuracy			0.6080	1000
macro avg	0.6090	0.6080	0.6075	1000
weighted avg	0.6090	0.6080	0.6075	1000

Why Baseline Failed: The DistilRoBERTa baseline (33.7%, barely above 33.3% random chance) was pre-trained on simpler NLI datasets (SNLI/MNLI) containing straightforward examples. On ANLI's adversarial examples, it exhibited extreme bias (85% neutral predictions), demonstrating that models trained on simpler tasks fail to generalize without adversarial fine-tuning.

Per-Class Performance (DeBERTa-v3-large)

Class	Precision	Recall	F1-Score	Support
Entailment	0.6201	0.6647	0.6416	333
Neutral	0.5940	0.5976	0.5958	332
Contradiction	0.6417	0.5916	0.6156	335

Analysis: DeBERTa achieves balanced performance across all classes (F1 range: 0.5958-0.6416) with no significant bias, successfully overcoming training set class imbalance through architecture and fine-tuning. Strong generalization confirmed by test accuracy (61.8%) exceeding validation accuracy (61.1%).

Hyperparameter Tuning Evidence

Experiment: Learning rate sensitivity analysis for DeBERTa-v3-large

Learning Rate	Dev Accuracy	Test Accuracy	Training Time
1e-5	58.2%	57.9%	46.5 min
2e-5	61.1%	61.8%	45 min
3e-5	59.7%	60.1%	45 min
5e-5	59.6%	59.95%	42.5 min

Optimal Configuration: 2e-5 learning rate with linear warmup (10% of steps), AdamW optimizer, 6 epochs, batch size 16 with gradient accumulation (effective batch size 32), gradient clipping at 1.0, and early stopping on validation F1. Mixed precision (FP16) training reduced memory by ~40% and improved speed by ~30% without loss of accuracy.

Evidence of Meaningfulness

Validation Through Stratified Sampling

To verify model quality beyond aggregate metrics, I performed stratified sampling: 2 samples per class from each data split (train/dev/test) yielding 18 representative samples with guaranteed coverage across all data distributions and label classes.

Representative Sample Performance: 28 correct / 8 incorrect = 77.8% accuracy (consistent with 61.8% full test set, accounting for small sample variance and higher difficulty in sampled adversarial examples).

Error Pattern Analysis

Primary Failure Mode (50% of errors): Over-inference on NEUTRAL cases

The model incorrectly infers definitive relationships from ambiguous information:

Premise: "San Pablo Reservoir is an open cut terminal water storage reservoir
owned by East Bay Municipal Utility District..."

Hypothesis: "The San Pablo Reservoir is man made"

True Label: NEUTRAL (not explicitly stated)

Predicted: ENTAILMENT (99.95% confidence)

→ Analysis: "Open cut" suggests but doesn't definitively indicate man-made construction; model over-inferred from contextual clues.

Secondary Issue (25% of errors): Numerical/Ordinal Reasoning

Premise: "...second-largest city in Kazakhstan, behind Almaty."

Hypothesis: "There were two other cities with higher population than Astana."

True Label: CONTRADICTION ("second-largest" means exactly ONE city is larger)

Predicted: ENTAILMENT (99.94% confidence)

→ Analysis: Model failed ordinal reasoning; "second" means position 2, implying only 1 city ranked higher, not 2.

Critical Observation: Model makes errors with very high confidence (90-99% on 7/8 errors), indicating overconfidence on edge cases requiring nuanced logical reasoning beyond pattern matching.

Success Pattern Analysis

The model correctly handles (28/36 samples, all >78% confidence):

Direct Contradictions:

Premise: "MGK released a mixtape in 2006... born April 22, 1990"

Hypothesis: "Released his first mixtape at age eighteen"

Predicted: CONTRADICTION (99.80% confidence) ✓

→ Correctly calculated: $2006 - 1990 = 16$ years old, not 18

Clear Entailments:

Premise: "Black Sabbath concluded 40+ year career"

Hypothesis: "Some bands have careers extending beyond 4 sets of ten years"

Predicted: ENTAILMENT (99.94% confidence) ✓

→ Correctly interpreted: 40+ years = 4+ decades

Obvious Neutral Cases: Correctly identified when hypotheses contain completely irrelevant information with no logical connection to premises.

Validation Against Benchmarks

- Generalization: Test accuracy (61.8%) exceeding dev accuracy (61.1%) confirms robust generalization without overfitting
- SOTA Comparison: Published ANLI R2 leaderboard shows 67-71% accuracy achieved through curriculum learning; our 61.8% is competitive for single-stage fine-tuning
- Architecture Validation: 28.1% improvement over baseline validates DeBERTa's architectural advantages for adversarial NLI

Conclusion: Stratified sampling reveals strong performance on clear-cut cases while identifying specific weaknesses (NEUTRAL/ENTAILMENT boundaries, numerical reasoning) that provide actionable directions for improvement through curriculum learning and confidence calibration.

Conclusions

This project successfully demonstrated a comprehensive approach to adversarial Natural Language Inference, achieving 61.8% accuracy on ANLI R2—a 28.1 percentage point improvement over baseline. The key achievements include: (1) systematic exploration from EDA through traditional ML to advanced transformers, validating that low word overlap (10-15%) necessitates semantic understanding beyond feature engineering, (2) identification that DeBERTa's disentangled attention provides 16.4% improvement over BERT specifically for adversarial examples, (3) demonstration that training set class imbalance (46.1% neutral) can be overcome through architectural design yielding balanced F1 scores (0.5958-0.6416), and (4) validation through stratified sampling revealing the model excels at clear cases but struggles with NEUTRAL/ENTAILMENT boundaries and numerical reasoning. The progression from 33.7% (baseline) to 38.7% (XGBoost) to 45.4% (BERT) to 61.8% (DeBERTa) demonstrates the necessity of both architectural innovation and adversarial-specific fine-tuning.

Proposed Next Steps:

1. Curriculum Learning Pipeline: Implement progressive pre-training on easier NLI datasets (SNLI → MNLI → FEVER-NLI) before final fine-tuning on ANLI R2. Based on published SOTA results (67-71%), this multi-stage approach is expected to yield 5-10 percentage point improvement, closing the gap to state-of-the-art. This directly addresses our finding that single-stage fine-tuning plateaus below optimal performance.
2. Confidence Calibration with Uncertainty Quantification: Address the critical issue of overconfident wrong predictions (90-99% confidence on errors) through temperature scaling, Monte Carlo dropout, or ensemble uncertainty estimation. This would improve model reliability on edge cases and provide actionable confidence thresholds for human-in-the-loop verification on ambiguous NEUTRAL/ENTAILMENT distinctions, where 50% of current errors occur.